CASE FILE

DarkSentinel

Darknet threat monitoring. Top 5 at the IIT Madras × BITS Goa hackathon.

PROBLEM

Monitoring darknet traffic needs a classifier, but sending every flow to a paid LLM would be costly and slow.

DECISIONS

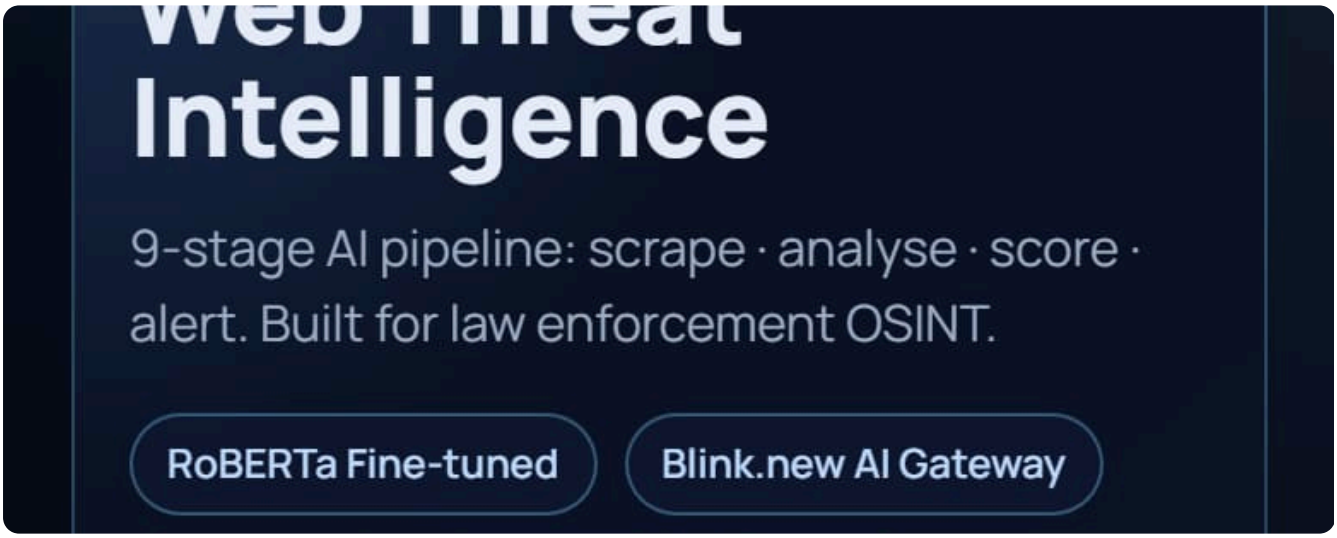
A Random Forest scores network flows for Tor traffic. Only predictions in the uncertain zone (Tor score 0.55 to 0.75) go to the Claude call, behind a 60-second cooldown, so cost is bounded by time rather than traffic. I removed a hardcoded fallback score that could show a result the model never produced.

RESULT

A team build (model training and live Tor capture were Deven's) with dashboards and an API; top 5 in its problem statement at the IIT Madras x BITS Goa hackathon 2026.

WHAT BROKE

A hardcoded fallback T-score could put an anomaly score on the dashboard that the model never produced, which made a missing prediction look like a working one. I removed the fallback from the analysis path.



Top 5 in problem statement, IIT Madras & BITS Goa hackathon 2026

Python · FastAPI · Next.js · Random Forest · RoBERTa · Claude API · Tor · Streamlit

---

dhruvkothari13062006@gmail.com · github.com/dhruv4740 · dhruvkothari.com